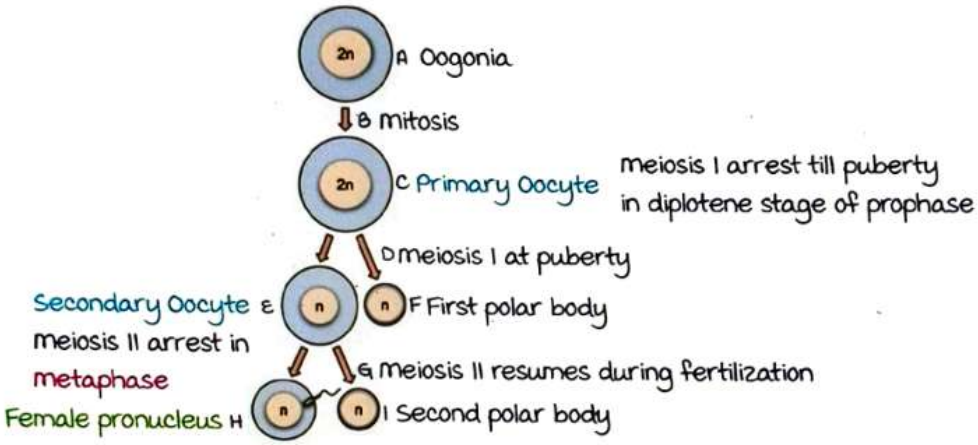
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FEMALE REPRODUCTION

----- Active space --

Oogenesis

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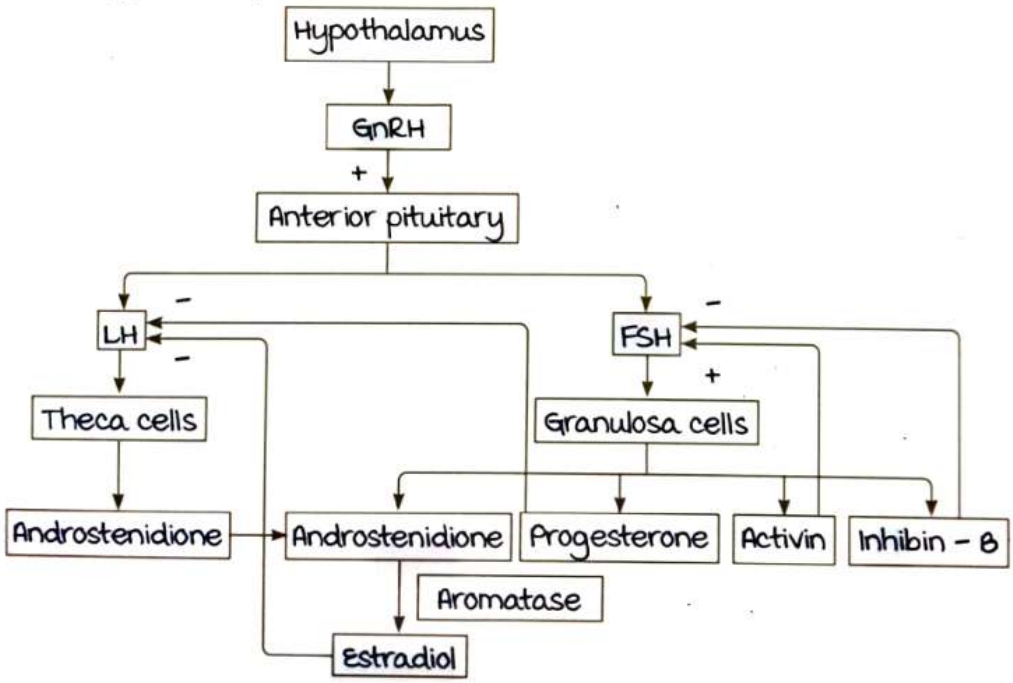


Regulation

00:03:50

Hypothalamic-pituitary-ovarian axis :

Two cell two gonadotropin model :



Menstrual cycle

00:07:53

28-day cycle in females.

Note :

Estrous cycle : Seen in mammals other than humans.

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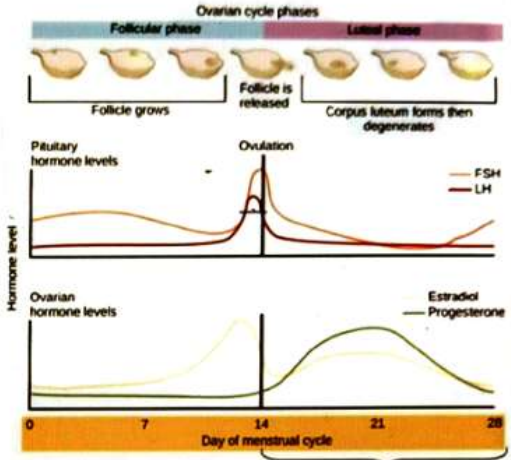
----- Active space -----

CHANGES IN OVARY :

Follicular phase :

FSH
↓
Follicular growth
↓
10–15 follicles grow → 1 follicle becomes dominant : Graafian follicle
- Maximum FSH receptors.
- Maximum Aromatase activity.
- Maximum estrogen production.

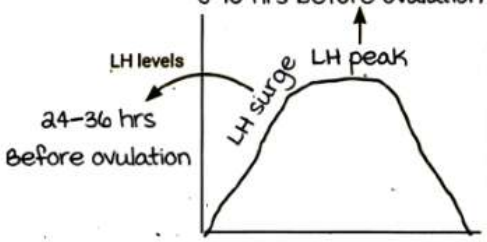
↓
Estrogen
↓ +feedback
↑↑ Luteinizing Hormone (LH surge)
↓
Acts on 1° oocyte → Meiosis I is completed → Ovulation
Progesterone synthesis
Synthesis of Prostaglandins



Ovarian changes in 28-day cycle

Constant duration (14d)

8–10 hrs before ovulation



FSH surge : Lysis of follicular wall on 14th day → Ovulation

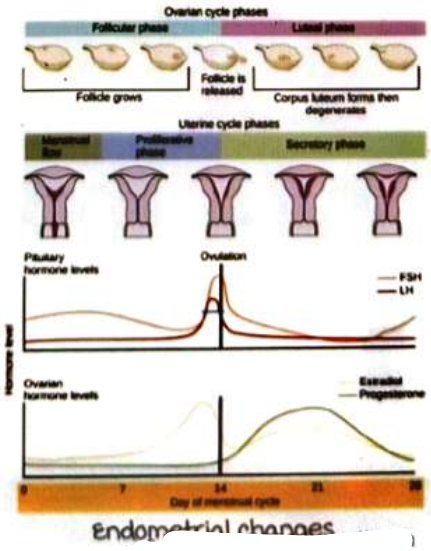
Luteal phase :

Bleeding inside Graafian follicle
↓
Clots
↓ LH
Luteal cells (Yellow-coloured)
↓
Corpus luteum → (If fertilization does not occur) → Degenerates on 24th day of cycle → menstrual blood loss
↓
Progesterone



Corpus luteum

CHANGES IN ENDOMETRIUM :

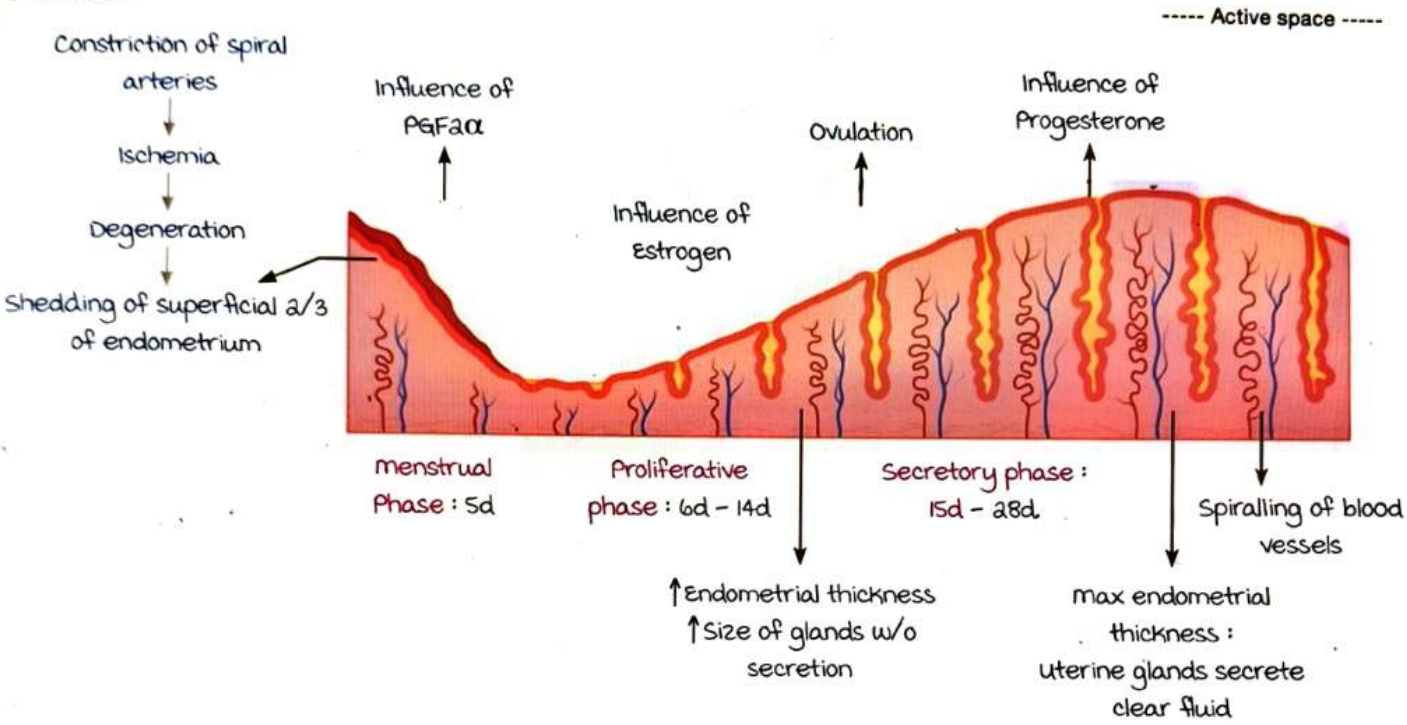


Endometrial changes

Feedback

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Phases :



CHANGES IN CERVIX :

Cervical mucus :

Under influence of

Estrogen

- Thin, watery (For easy sperm movement).
- Spinnbarkeit effect : Stretchable (Elasticity).
- ↑ NaCl & ↓ proteins → Fern pattern.


cervical mucus


fern pattern

Progesterone :

- Thick & less elastic pattern.

CHANGES IN VAGINA

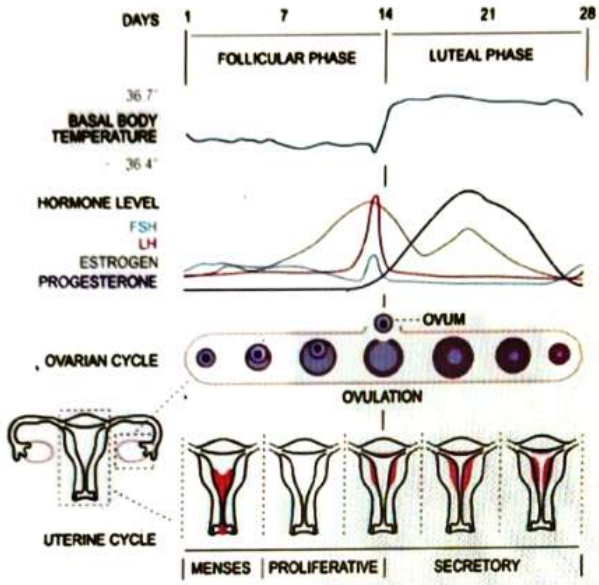
Parabasal cells :	Superficial cells :	Intermediate cells :
Lack of hormonal activity	Dominate under estrogen influence	Dominate under progesterone influence

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----- Active space -----

SUMMARY :



Basal Body Temperature :

- ↑ by 0.5–1°C during Ovulation.
- D/t to progesterone (Thermogenic action).

menstrual cells

Female Sex Hormones

00:34:56

Steroid Hormones

- Estrogen.
- Progesterone.

Protein Hormones

- Inhibin.
- Activin.
- Relaxin.

STEROID HORMONES :

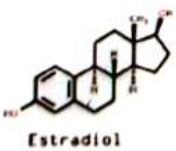
Estrogen :

C-18 steroid.

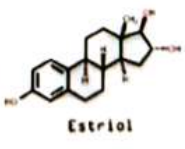
Sources : Granulosa cells, Placenta, Corpus luteum.

Forms :

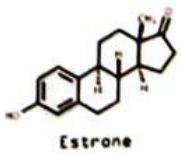
Estradiol : Reproductive age group



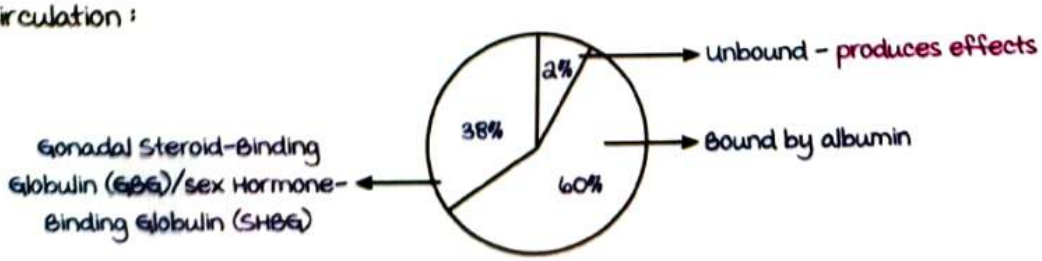
Estriol : Pregnancy



Estrone : menopause



In circulation :



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Actions :

----- Active space -----

Uterus (Estrogen dominant)	• ↑ Blood flow. • more active : ↑ Contractility & ↑ Excitability.
Secondary sexual characteristics	• Female body contour. • Fat distribution in breast & buttocks.
Breast	• Growth promotion. • Ductal proliferation.
Bone	• Produces Osteoprotegerin (inhibitor of RANK ligand) → ↓osteoclast differentiation → inhibition of bone resorption.
Liver	• ↓ Plasma LDL cholesterol.
CVS	• Inhibits platelet activation. • Promotes vasodilation by ↑ NO.
CNS	• Neuroprotective : inhibits neuronal cell death.
Kidneys	• Salt & water retention.

Progesterone :

Actions :

	Anti-estrogenic action : less active, less excitable, less contractile.
Breast	• Promotes lobulo-alveolar growth
CNS	• Thermogenic action : ↑ in BBT by 0.5-1°C during ovulation.
Respiration	• Stimulates respiration → CO_2 washout → ↓ in alveolar PCO_2 levels.
Kidneys	• Natriuresis : salt & water excretion. (only steroid hormone to cause natriuresis).

PROTEIN HORMONES :

Inhibin :

Sources :

- Corpus luteum : Inhibin A
- Granulosa cells : Inhibin B

Action : Negative feedback inhibition of FSH

Relaxin :

Sources : Uterus, Placenta.

Actions :

- Relaxes pubic symphysis
- Dilates cervix

} Aids in labour

Activin : Stimulates FSH.

Follistatin : Inhibits FSH.

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----- Active space -----

HUMAN CHORIONIC GONADOTROPIN (HCG):

Glycoprotein hormone.

Source: Syncytiotrophoblast.

Subunits:
- α-subunit: Similar to FSH, LH, TSH.
- β-subunit: Similar to LH.

	hCG	LH
T1/2	37 hours	30 minutes
Potency	80x	x

Functions:

- Maintenance of corpus luteum up to 6 weeks of pregnancy
- Leydig cells → Testosterone
- Immunosuppressive activity → No fetal rejection
- Corpus luteum → Relaxin

Lactation

00:54:27

Processes seen:

	Meaning	Hormones involved
1. Mammogenesis	Growth of breast	• Estrogen: Ductal growth. • Progesterone: Lobulo-alveolar growth.
2. Lactogenesis	Production of milk.	• Prolactin. • Insulin. • Growth Hormone.
3. Galactokinesis	Release of milk.	Milk ejection/let down reflex: • Suckling → Stimulates hypothalamus → Oxytocin release from posterior pituitary. • Neuroendocrine reflex. • Inhibited by anxiety, pain, depression.
4. Galactopoiesis	Maintenance of lactation.	• Prolactin

Feedback